Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	**Aim: To obtain the classification of various classes using k-Nearest Neighbor approach**	
Experiment No: 03	**Date: 04-02-2026**	**Enrollment No: 92410133023**

Aim: To obtain the classification of various classes using k-Nearest Neighbor approach

IDE: Google Colab

Theory:

This algorithm is used to solve the classification model problems. K-nearest neighbor or K-NN algorithm basically creates an imaginary boundary to classify the data. When new data points come in, the algorithm will try to predict that to the nearest of the boundary line. Therefore, larger k value means smother curves of separation resulting in less complex models. Whereas, smaller k value tends to over fit the data and resulting in complex models. It's very important to have the right k-value when analyzing the dataset to avoid over fitting and under fitting of the dataset.

The model representation for KNN is the entire training dataset. It is as simple as that. KNN has no model other than storing the entire dataset, so there is no learning required. Efficient implementations can store the data using complex data structures like k-d trees to make look-up and matching of new patterns during prediction efficient. Because the entire training dataset is stored, you may want to think carefully about the consistency of your training data. It might be a good idea to curate it, update it often as new data becomes available and remove erroneous and outlier data.

K-NN algorithm assumes the similarity between the new case/data and available cases and put the new case into the category that is most similar to the available categories. K-NN algorithm stores all the available data and classifies a new data point based on the similarity. This means when new data appears then it can be easily classified into a well suite category by using K- NN algorithm. K-NN algorithm can be used for Regression as well as for Classification but mostly it is used for the Classification problems.

K-NN is a non-parametric algorithm, which means it does not make any assumption on underlying data. It is also called a lazy learner algorithm because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset. KNN algorithm at the training phase just stores the dataset and when it gets new data, then it classifies that data into a category that is much similar to the new data.

Example: Suppose, we have an image of a creature that looks similar to cat and dog, but we want to know either it is a cat or dog. So for this identification, we can use the KNN algorithm, as it works on a similarity measure. Our KNN model will find the similar features of the new data set to the cats and dogs images and based on the most similar features it will put it in either cat or dog category.

Methodology:

1. Load the basic libraries and packages
2. Load the dataset
3. Analyse the dataset
4. Pre-process the data

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5. Visualize the Data
6. Separate the feature and prediction value columns
7. Select the number K of the neighbors
8. Calculate the Euclidean distance of K number of neighbors
9. Take the K nearest neighbors as per the calculated Euclidean distance.
10. Among these k neighbors, count the number of the data points in each category.
11. Assign the new data points to that category for which the number of the neighbor is maximum.
12. Our model is ready.

Pre Lab Exercise:

a. What is K in KNN-approach?
   K represents the number of nearest neighbors considered to classify or predict a data point.

b. Which are the types of distance metrices?
   Common distance metrics used in KNN are:
   - **Euclidean Distance**
   - **Manhattan Distance**
   - **Minkowski Distance**
   - **Hamming Distance** (for categorical data)

c. What can be the criteria of selection of the value of K?

   - Choose **odd value of K** (to avoid ties)
   - Use **small K** for less noise, **large K** for smoother results
   - Select K using **cross-validation**
   - Rule of thumb: K ≈ $\sqrt{n}$ (n = number of data points)

d. What are the advantages of KNN algorithm?
   - Simple and easy to understand
   - No training phase(lazy learning)
   - Works well with small datasets
   - Can handle both classification and regression

Program (Code):

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import confusion_matrix, accuracy_score
from sklearn.preprocessing import StandardScaler

# Sample dataset (replace with your dataset)
from sklearn.datasets import load_iris
data = load_iris()
X = data.data
y = data.target

# Train test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Feature scaling
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)

# Model
model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)
```

```
# Results
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("Accuracy Score:", accuracy_score(y_test, y_pred))
```

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Results:

To be attached with

a. Confusion matrix

```
Confusion Matrix:
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

b. Accuracy Score

```
Accuracy Score: 1.0
```

Observation and Result Analysis:

a. Nature of the dataset

It was observed that the dataset plays a very important role in model performance. A properly structured and balanced dataset helps the model learn patterns effectively. If the dataset contains noise, missing values, or is imbalanced, the model accuracy may decrease. Therefore, good quality data is essential for better results.

b. During Training Process

During the training process, the model learns patterns from the data by adjusting its parameters. It was observed that the model performance improves as it is trained with more data. Proper selection of parameters (like value of K in KNN) is important, as incorrect values can lead to poor performance. The model gradually reduces error during training.

c. After the training Process

After training, the model is tested on unseen data to evaluate its performance. It was observed that if the model performs well on both training and testing data, it means the model has learned correctly. If there is a large difference between training and testing performance, it indicates overfitting or underfitting.

d. Observation over the Learning Curve

From the learning curve, it was observed that the model performance changes as training progresses. If both training and validation errors decrease and become stable, the model is performing well. If there is a large gap between them, it indicates overfitting. If both errors remain high, it indicates underfitting. Thus, the learning curve helps in understanding model behaviour.

Post Lab Exercise:

a. Is KNN useful in regression-based problems?

Yes, KNN can be used for regression problems. In KNN regression, the output is calculated as the average (or weighted average) of the values of the K nearest neighbors. Instead of predicting a class label, it predicts a continuous value. Therefore, KNN is useful for both classification and regression tasks.

b. Why KNN is a non-parametric algorithm?

KNN is called a non-parametric algorithm because it does not assume any underlying distribution or form of the data. It does not build a mathematical model during training. Instead, it directly uses the training data to make predictions. Hence, it is flexible and can work with different types of data.

c. What are the assumptions of KNN approach?

KNN assumes that similar data points are located close to each other in the feature space. It assumes that distance (such as Euclidean distance) is a meaningful measure of similarity. It also assumes that all features contribute equally unless properly scaled. Therefore, feature scaling is important in KNN.

d. How can the KNN approach make the prediction of the unseen dataset?

KNN makes predictions by finding the K nearest neighbors of a new (unseen) data point based on a distance metric. For classification, it assigns the most common class among the neighbors. For regression, it calculates the average value of the neighbors. Thus, prediction is based on similarity with existing data points.

e. Why is KNN called a Lazy Learner?

KNN is called a lazy learner because it does not learn or build a model during the training phase. It simply stores the training data. The actual computation happens only when a new data point needs to be predicted. This delays learning until prediction time, hence the name “lazy learner”.

f. Why should we not use KNN for large dataset?

KNN is not suitable for large datasets because it requires calculating the distance between the new data point and all training data points, which is computationally expensive. This increases prediction time and memory usage. As the dataset grows, KNN becomes slower and less efficient.